

File Edit Format Run Options Window Help

```
import datetime
picdir = os.path.join(os.path.dirname(os.path.realpath(__file__)), 'pic')
libdir = os.path.join(os.path.dirname(os.path.realpath(__file__)), 'lib')
if os.path.exists(libdir):
    sys.path.append(libdir)

import logging
import time
import traceback
from waveshare_OLED import OLED_1in5
from PIL import Image, ImageDraw, ImageFont
logging.basicConfig(level=logging.DEBUG)

try:
    disp = OLED_1in5.OLED_1in5()
    x = datetime.datetime.now()
```

Fig. 4: A snapshot of the code

File Edit Format Run Options Window Help

```
draw = ImageDraw.Draw(image1)
font = ImageFont.truetype(os.path.join(picdir, 'Font.ttc'), 12)
font1 = ImageFont.truetype(os.path.join(picdir, 'Font.ttc'), 18)
font2 = ImageFont.truetype(os.path.join(picdir, 'Font.ttc'), 24)
logging.info ("***draw line")
draw.line([(0,0),(127,0)], fill = 15)
draw.line([(0,0),(0,127)], fill = 15)
draw.line([(0,127),(127,127)], fill = 15)
draw.line([(127,0),(127,127)], fill = 15)
logging.info ("***draw text")
draw.text((20,0), str(x), font = font1, fill = 15)

image1 = image1.rotate(0)
disp.ShowImage(disp.getbuffer(image1))
time.sleep(3)

logging.info ("***draw rectangle")
image1 = Image.new('L', (disp.width, disp.height), 0)
draw = ImageDraw.Draw(image1)
for i in range(0, 16):
    draw.rectangle([(0, 8*i), (128, 8*(i+1))], fill = 1)
disp.ShowImage(disp.getbuffer(image1))
time.sleep(3)

logging.info ("***draw image")
Himage2 = Image.new('L', (disp.width, disp.height), 0) # 0: clear the frame
bmp = Image.open(os.path.join(picdir, '1in5.bmp'))
Himage2.paste(bmp, (0,0))
Himage2=Himage2.rotate(0)
```

Fig. 5: Another snapshot of the code

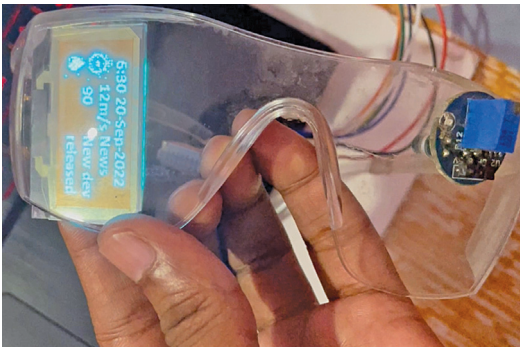


Fig. 6: Final prototype used for testing

```
wget http://www.airspayce.com/
mikem/bcm2835/bcm2835-1.71.tar.
gz
tar zxvf bcm2835-1.71.tar.gz
cd bcm2835-1.71/
sudo ./configure && sudo make &&
sudo make check && sudo make
install
sudo apt-get install wiringpi
```

#For Raspberry Pi systems after May 2019 (earlier than this, you may not need execute), you may need to

upgrade:

```
wget https://project-downloads.drogon.net/wiringpi-latest.deb
```

```
sudo dpkg -i wiringpi-latest.deb
```

Check the installation using the following:

```
gpio -v
```

Install python modules:

```
sudo apt-get install python3-pil
```

```
sudo apt-get install python3-numpy
```

```
sudo pip3 install RPi.GPIO
```

```
sudo pip3 install spidev
```

Next, clone the transparent OLED Python library and install the library:

```
sudo apt-get install p7zip-full
```

```
sudo wget https://www.waveshare.com/w/
```

```
upload/2/2c/OLED_Module_Code.7z
```

```
7z x OLED_Module_Code.7z -O./OLED_
```

```
Module_Code
```

```
cd OLED_Module_Code/RaspberryPi
```

To start coding, first import the Python modules and libraries. For the prototype, the DateTime library, OLED display modules and libraries were imported. You can also import many other libraries and modules, if you want to add any more functions to the smart eyeglasses.

Note. You can download the source code from following link: <https://www.electronicsforu.com/electronics-projects/smart-google-glasses-with-transparent-oled-display>

Construction and testing

Now we have our smart glasses ready to be used. On powering the Raspberry Pi and running the code, you can see the information being displayed on the transparent OLED, like date and time, and other information that you have set in the code to be displayed. The information will appear as an overlay or an image floating in the air, which you may have seen in some science fiction movies.

Author's final prototype used for testing is shown in Fig. 6. This is the first version of the project. In next version, we shall try to add features like controlling the display using eyes and getting notifications from the phone. **EFY**

Ashwini Kumar Sinha is a technology enthusiast at EFY

is given here. To install the Python modules, open the Linux terminal and run the following command: